Machine Learning - Undergraduate Level Assessment 13 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Consider the Bayesian network given below. How many independent parameters would we need if we made no assumptions about independence or conditional independence H -> U <- P <- W?
   (a) 3
   (b) 4
   (c) 7
   (d) 15
2. If N is the number of instances in the training dataset, nearest neighbors has a classification run time of
   (a) O(1)
   (b) O(N)
   (c) O(log N)
   (d) O(N^2)
3. Which one of the following is the main reason for pruning a Decision Tree?
   (a) To save computing time during testing
   (b) To save space for storing the Decision Tree
   (c) To make the training set error smaller
   (d) To avoid overfitting the training set
4. Statement 1| L 2 regularization of linear models tends to make models more sparse than L 1 regularization. Statement 2| Residual connections can be found in ResNets and Transformers.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True
5. Statement 1| The softmax function is commonly used in mutliclass logistic regression. Statement 2| The temperature of a nonuniform softmax distribution affects its entropy.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: The computational complexity of gradient descent is linear in the number of data points, N.
7. True or False: The tanh function is a linear activation function commonly used in neural networks.
8. True or False: Averaging the output of multiple decision trees helps increase bias in the model's predictions.
9. True or False: The softmax function is indeed commonly used in multiclass logistic regression to model probability distributions over classes.
10. True or False: The BLEU metric evaluates translation quality using precision, whereas the ROGUE metric focuses on recall for summarization tasks.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the longest chain which can be formed from the given set of pairs.
12. Write a function to find maximum possible sum of disjoint pairs for the given array of integers and a number k.

End of Paper